Compact Fixed Mount 2D scanner

mars

MCR-F180

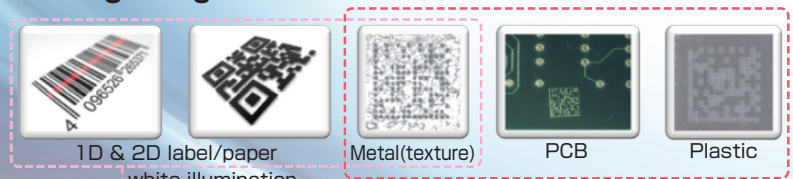
High speed & consistent DPM scanning with Innovative Illumination!
Configuration tool is ready to ease the use!

※DPM=Direct Parts Marking

Innovative new design!

Combination of two different types of illumination

Select the best combination of white & red high brightness LEDs





※DPM=Direct Parts Marking



MCR-F180

1D symbol | 2D symbol | LAN | Direct Marking | PoE | DI/O | PLC-link | High resolution | IP67

- **White illumination**
  Less reflection like label/paper
- **Red illumination**
  Reflective, less contrast, DPM

Strength

High quality 1.2M pixel CMOS

For high speed reading and small object

- **Minimum resolution: 0.125mm**
- **Working distance: 35 to 170mm**

■Depth of field : 1D symbol

Narrow bar width	Near	Far
0.125	60	120
0.15	55	130
0.19	40	140
0.25	30	145
0.33	25	175

※Under the condition of using our standard symbols (unit:mm)

■Depth of field : 2D symbol

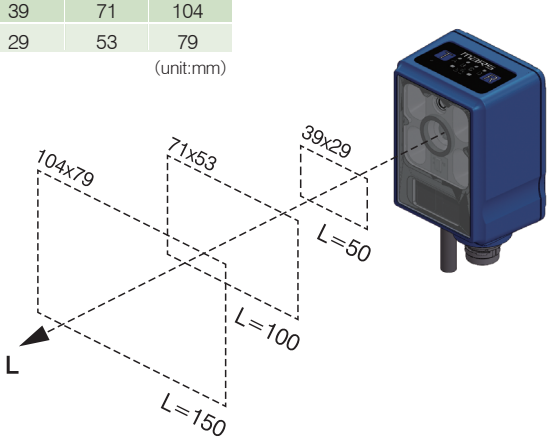
Cell size	Near	Far
0.125	65	75
0.167	65	115
0.19	65	125
0.25	60	135
0.33	45	135

※Under the condition of using our standard symbols (unit:mm)

■Field of View

Distance	50	100	150
Horizontal	39	71	104
Vertical	29	53	79

(unit:mm)

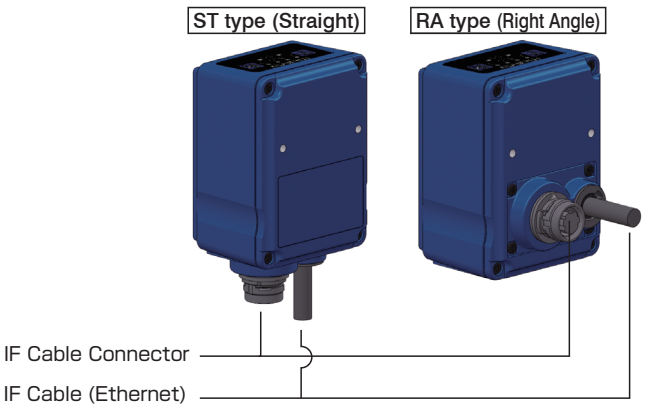


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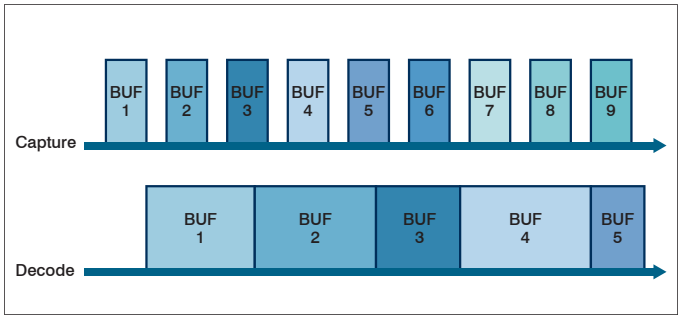
Flexible with space limitation

Types of cable angle are available.



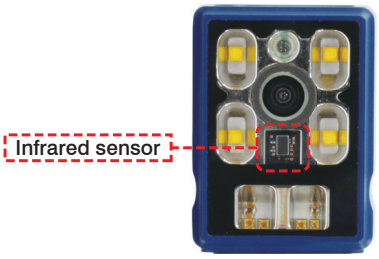
Read moving object with Cycle Buffer function

MCR-F180 can capture images continuously with the minimum interval and attempt to read the images in parallel. The image buffer is maximum for 64 images, which should be enough for the most of applications.



Proximity sensor to detect object

MCR-F180 employed Infrared sensor to detect object to read with high accuracy.



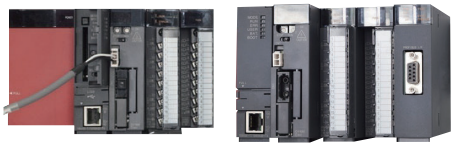
Easy setting

Updated cofiguration tool "MCRConfig2" is available.



PLC Link

Set the memory address of PLC for barcode data into the scanner and then MCR-F180 writes the reading data into the memory of PLC automatically. Only the memory monitoring will be needed to use MCR-F180 with your PLC. Read/Write data accompanies with hand shake by using PLC memory.



General specification

Model name		MCR-F180-ST/MCR-F180-RA	MCR-F180P-ST/MCR-F180P-RA
Receiver	Sensor	1/3 Inches CMOS Image Sensor Monochrome	
	Pixels	1280(X)×960(Y) (1.2 million pixels)	
Light emission	Illumination	High-intensity white & Red LED	
	Pointer	Blue LED	
Supported symbol	1D	Codabar,Code39,Code128/GS1-128,Code93,GS1 DataBar,ITF,AztecCode,PDF417/	
	2D	MicroPDF,CompositeCode,UPC/EAN/JAN,Data Matrix (ECC200) , QR Code/Micro QR	
Angle of Reading		PITCH:±35°/ SKEW: ±35°/ TILT: 360°	
Minimum resolution		0.125 mm	
Depth of field		35～170 mm	
FOV at 100mm distance		71 x 53 mm	
Power Rating	Supply Voltage	DC24V±10% ※Can not use DC and PoE simultaneously.	PoE Type A ※Can not use DC and PoE simultaneously.
	Current Consumption	Command (SYNC) standby: 150mA During reading operation: 400mA	PoE Power Class3

Model name		MCR-F180-ST/MCR-F180-RA	MCR-F180P-ST/MCR-F180P-RA
Proximity sensor		Semiconductor laser 940nm Class 1	
Digital input		Photo-coupler isolation x 2, Input resistance 2.2kOhm	
		OFF voltage 0 - 0.8V, ON voltage 10.8 to 28V	
Digital output		Photo-coupler isolation x 2, Maximum rating DC30V 50mA	
Interface		Ethernet（10Base-T、100Base-TX）	
		Length of Cable 2m (LAN Connection)	
Environmental	Operating Temparature	0 ～ 40℃	
	Operating Humidity	35 ～ 85%R.H.(no condensing)	
	Storage Temparature	-20 ～ 60℃	
	Storage Humidity	35 ～ 85%R.H.(no condensing)	
	Vibration	10-55Hz amplitude1.5mm(full), 2hours for X,Y,Z each direction	
	Seal Rating	IP67	
	Ambient Light	less than 10000 lx	
Dimensions		60(H) x 45(W) x 35(D) Unit : mm (Contain no connector protrusion)	
Weight		Approx 220g	Approx 235g

※The above specs are the subject to be changed without notification

Contact

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ISO 9001 Obtained certification

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